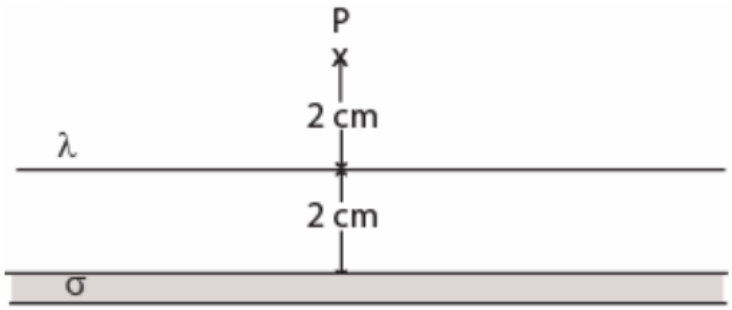
A proton enters a region of uniform electric field $\vec{E} = 80\ \hat{j}$ (N/C) with an initial velocity $\vec{v} = 20\ \hat{i}$ (km/s). What is the speed of the proton 2.0 μs after entering this region?

A) 25 km/s
B) 15 km/s
C) 42 km/s
D) 35 km/s
E) 4.7 km/s

Consider a conducting neutral spherical shell having an inner radius of 3.70 cm and an outer radius of 4.50 cm. A positive point charge q is placed at the center of the shell. The magnitude of the electric field a distance 5.00 cm from the center of the shell is 2500 N/C. Calculate the magnitude of the charge density on the outer surface of the shell.

A) 2.73×10^{-8} C/m^2
B) 1.34×10^{-8} C/m^2
C) 5.16×10^{-8} C/m^2
D) 6.10×10^{-7} C/m^2

Consider an infinitely large non-conducting flat sheet carrying a uniform charge density σ = +20 nC/m^2 and a long thin wire carrying a uniform charge density λ = −2.0 nC/m arranged as shown in figure . The magnitude of the net electric field due to these two charge distributions at point P is



A) 670 N/C
B) 450 N/C
C) 240 N/C
D) 120 N/C
E) 930 N/C

Two identical positively charged ions are separated from each other by a distance of 6.8×10^{-9} m. If the electrostatic force between them is 4.5×10^{-9} N, how many electrons are missing from each ion?

A) 30
B) 45
C) 48
D) 37
E) 25

Three electric charges $Q_A = Q_B = q$, and $Q_C = -2q$ are located a the points A ($x = +a$, $y = 0$), B ($x = -a$, $y = 0$), and C ($x = 0$, $y = +2a$), respectively. What is the electric field at the origin?

A) $k\frac{q}{2a^2}$ toward Q_C

B) $k\frac{q}{2a^2}$ away from Q_C

C) $k\frac{q}{a}$ toward Q_C

D) $k\frac{q}{a}$ away from Q_C

E) zero

A large insulating solid sphere has a charge density of 5 nC/m^3. Calculate the electric field inside the sphere at a distance of 10 cm from its center.

A) 12.6 N/C
B) 0
C) 26.4 N/C
D) 18.8 N/C
E) 5.50 N/C

A long solid non-conducting cylinder (radius = 12 cm) has a uniform charge density (5.0 nC/m^3) distributed throughout its volume. Determine the magnitude of the electric field 5.0 cm from the axis of the cylinder.

A) 5 N/C
B) 14 N/C
C) 31 N/C
D) 25 N/C
E) 20 N/C

Two conducting spheres, one having twice the diameter of the other, are separated by a distance large compared to their diameters. The smaller sphere has charge q and the larger sphere is uncharged. If the spheres are connected by a long thin conducting wire:

A) 1 and 2 have the same charge
B) The value of the electric field at both surfaces is same
C) 1 and 2 have the same potential
D) 2 has half the potential as 1
E) 2 has twice the potential as 1

Three large insulating sheets of charge with the given charge densities are shown in figure . The magnitudes of electric field at points A and B are respectively

A) $3\sigma_o / \varepsilon_o$, $3\sigma_o / \varepsilon_o$
B) $2\sigma_o / \varepsilon_o$, 0
C) $\sigma_o / 2\varepsilon_o$, $\sigma_o / 2\varepsilon_o$
D) $3\sigma_o / \varepsilon_o$, 0
E) σ_o / ε_o , 0

